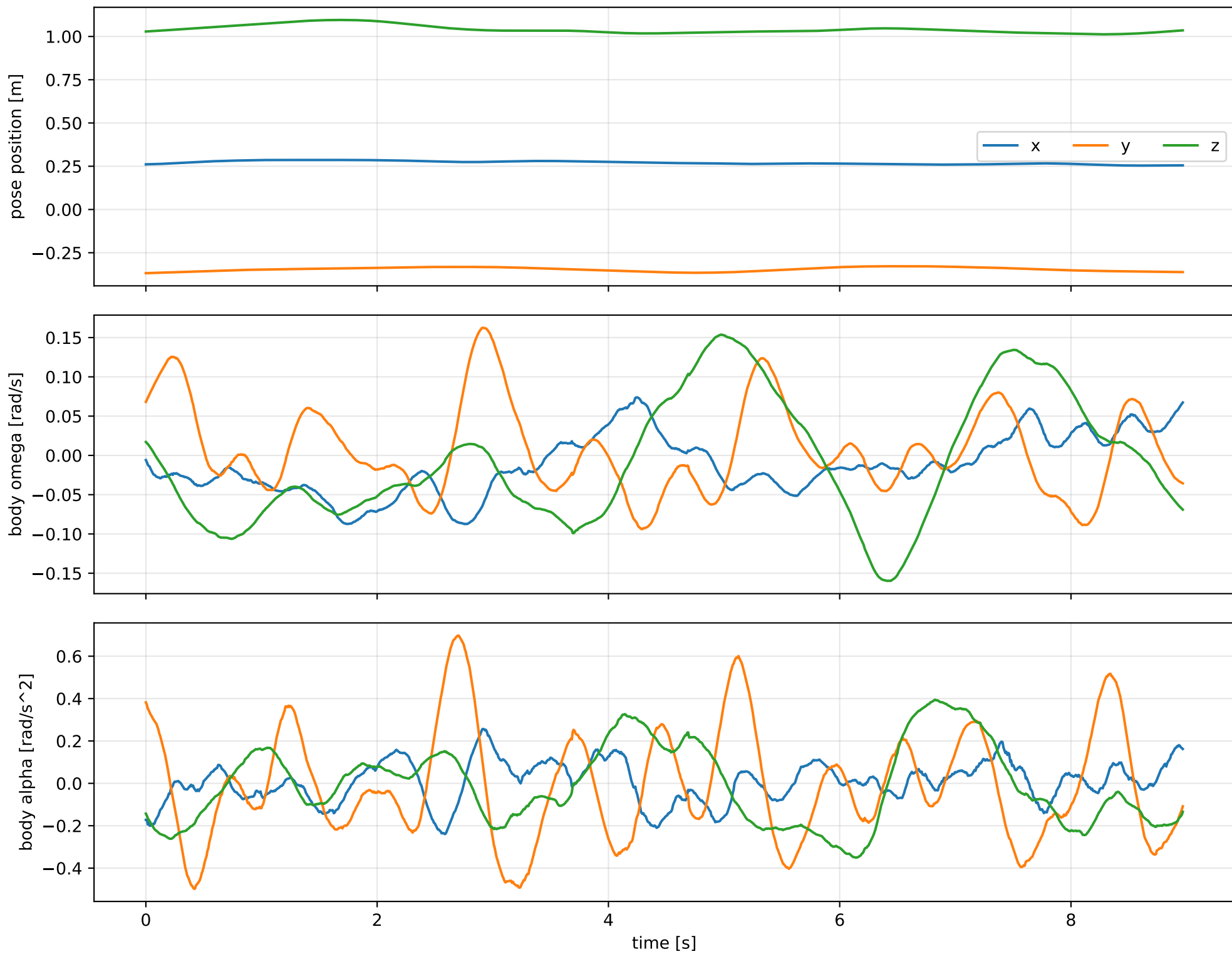
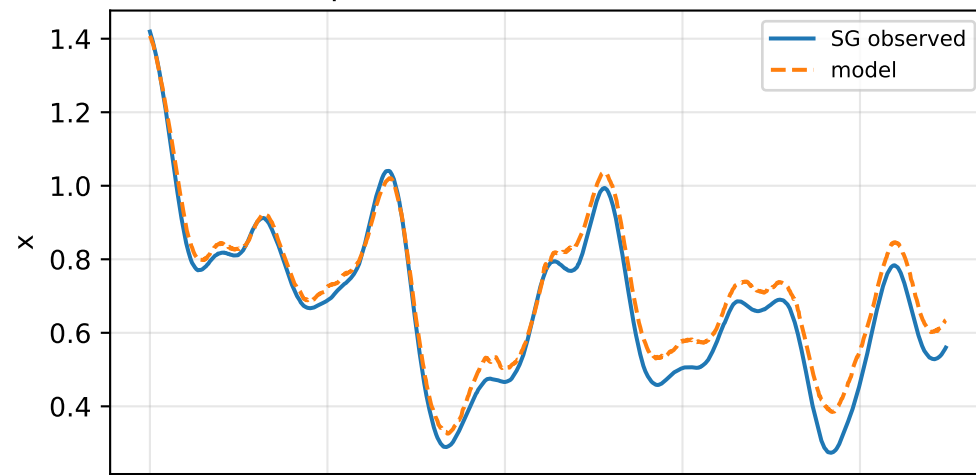


kkt_scale_positive: SG trajectory and derived motion

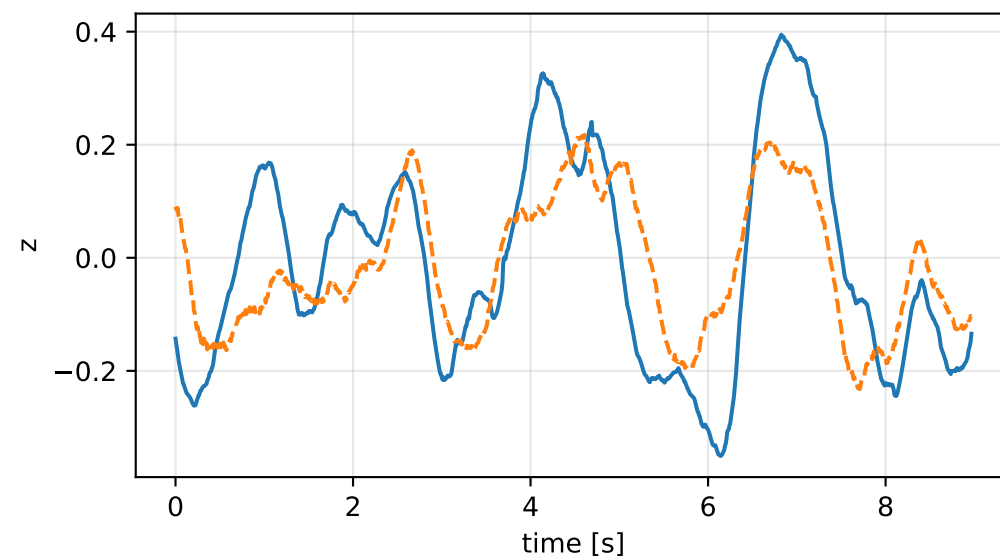
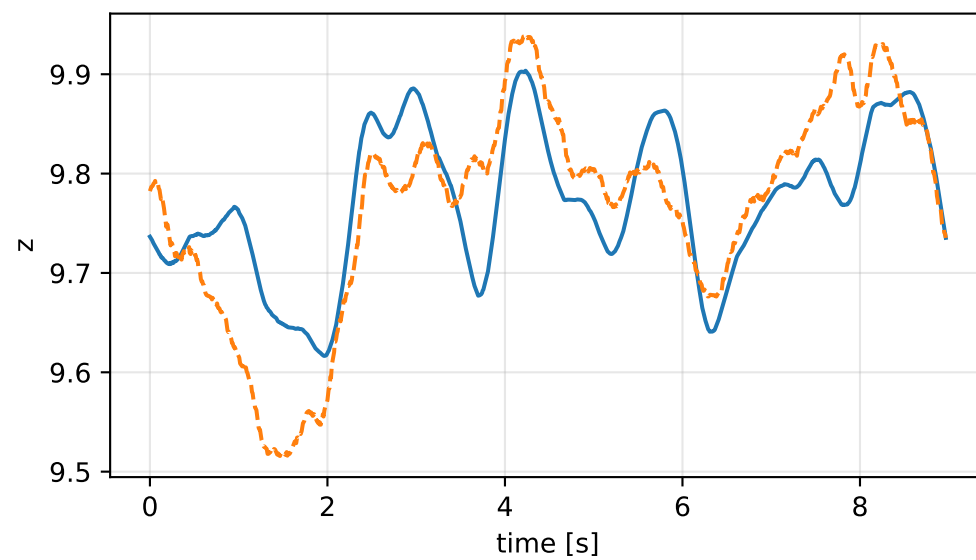
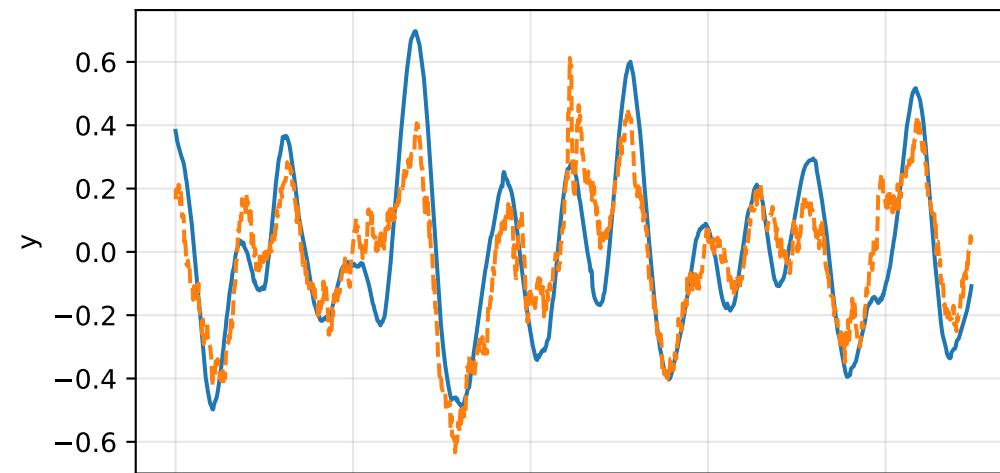
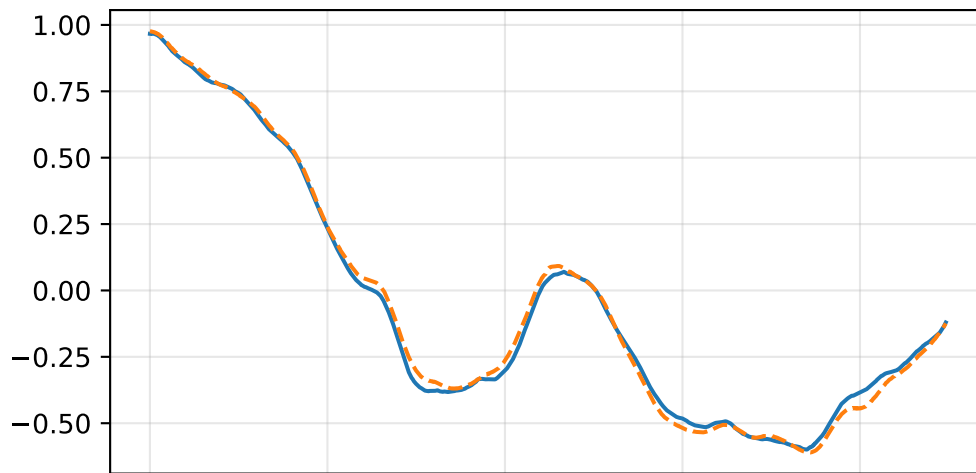
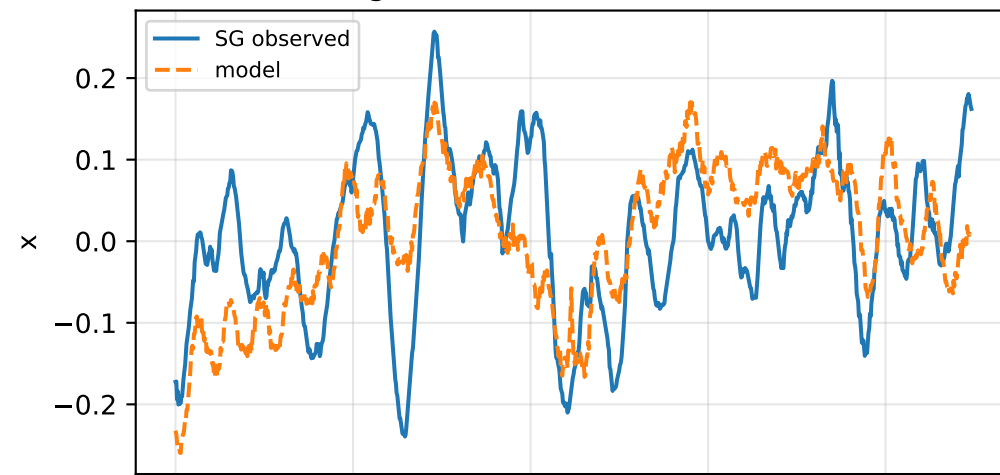


kkt_scale_positive: acceleration residual objective

specific acceleration [m/s^2]

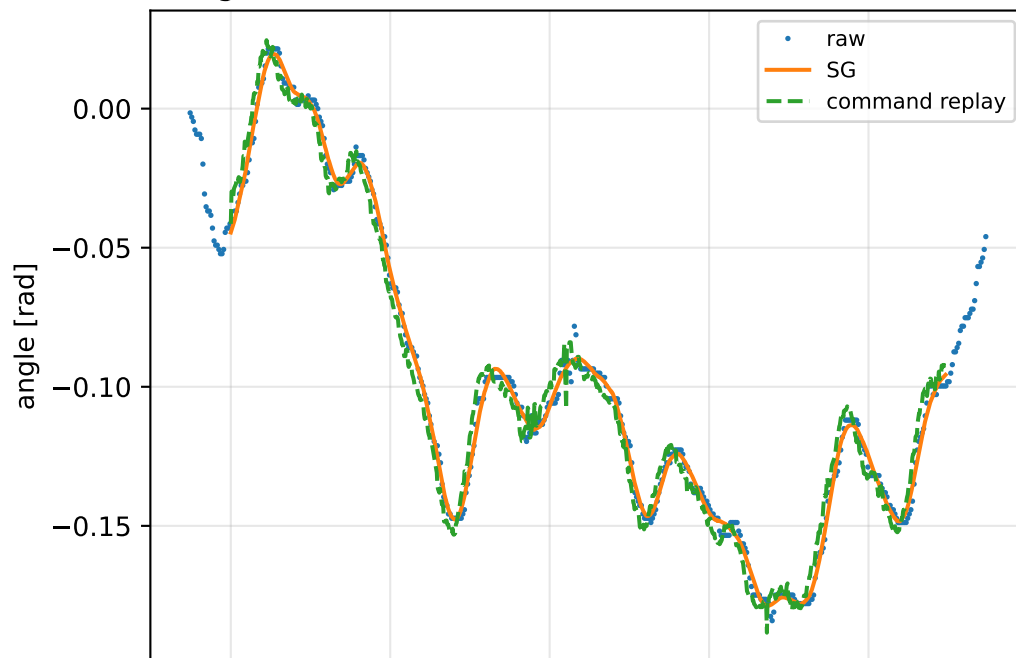


angular acceleration [rad/s^2]

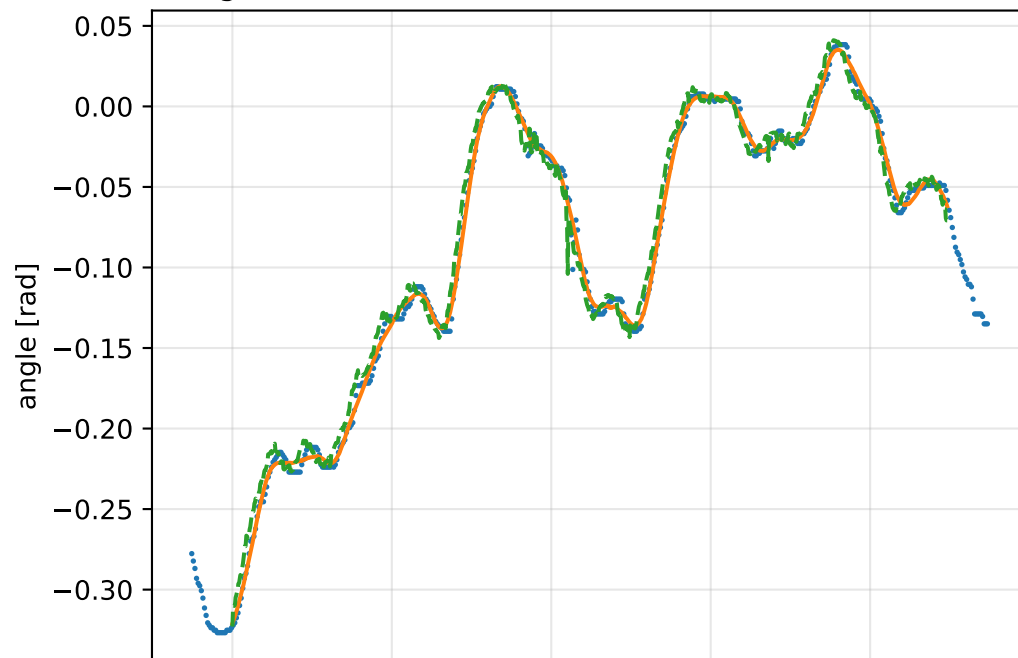


kkt_scale_positive: gimbal measurement audit (objective=measured_sg)

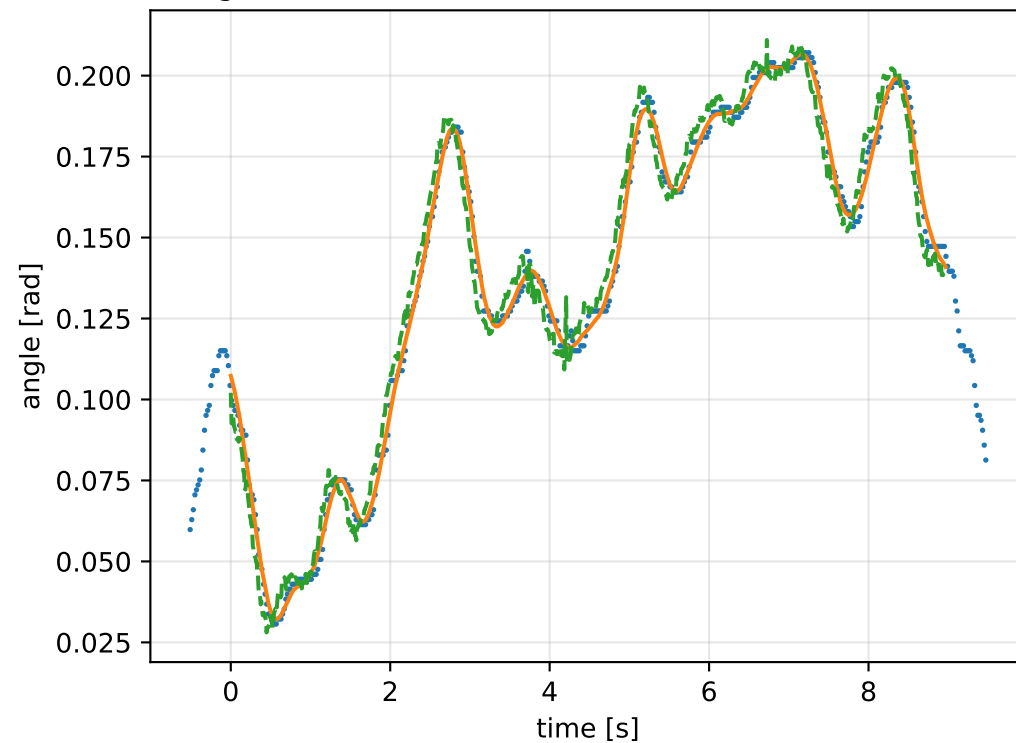
gimbal 1: RMSE=0.006345, max=0.01494 rad



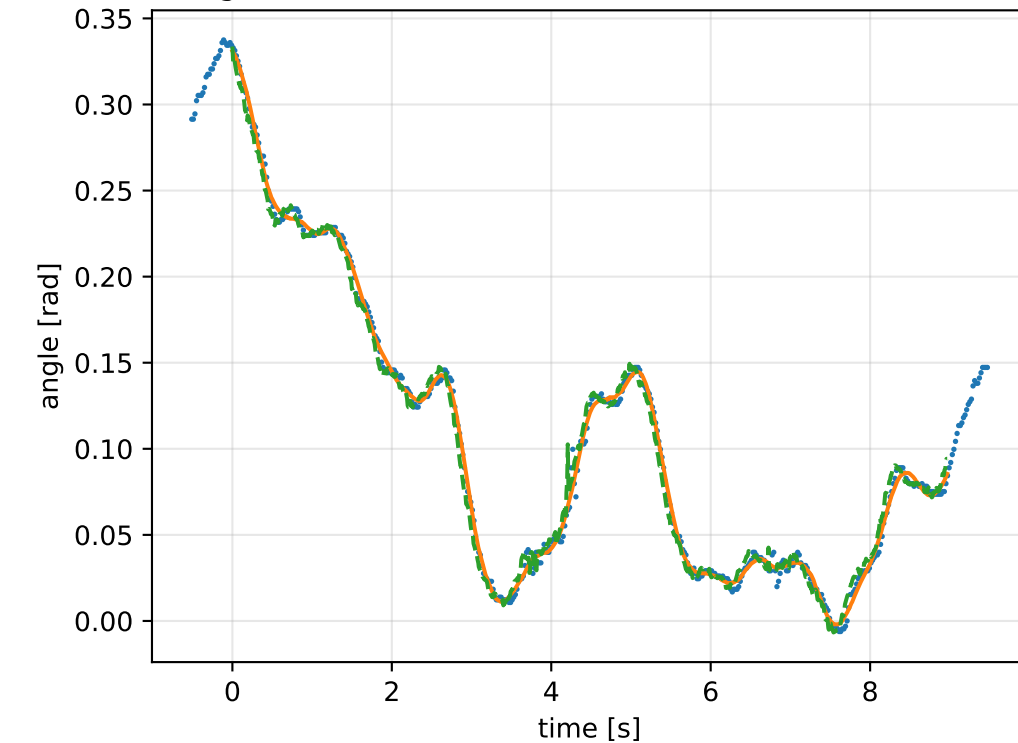
gimbal 2: RMSE=0.008835, max=0.04462 rad



gimbal 3: RMSE=0.007019, max=0.01611 rad

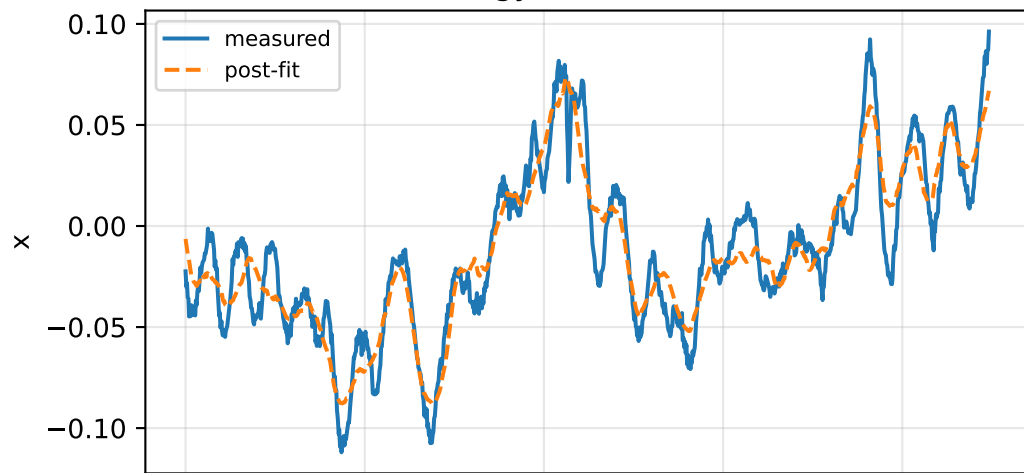


gimbal 4: RMSE=0.006321, max=0.03642 rad

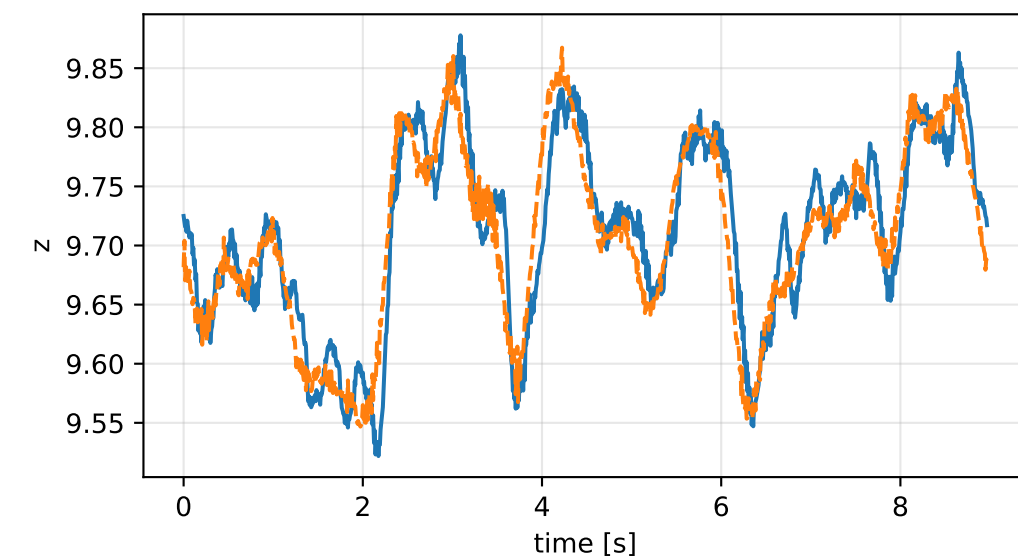
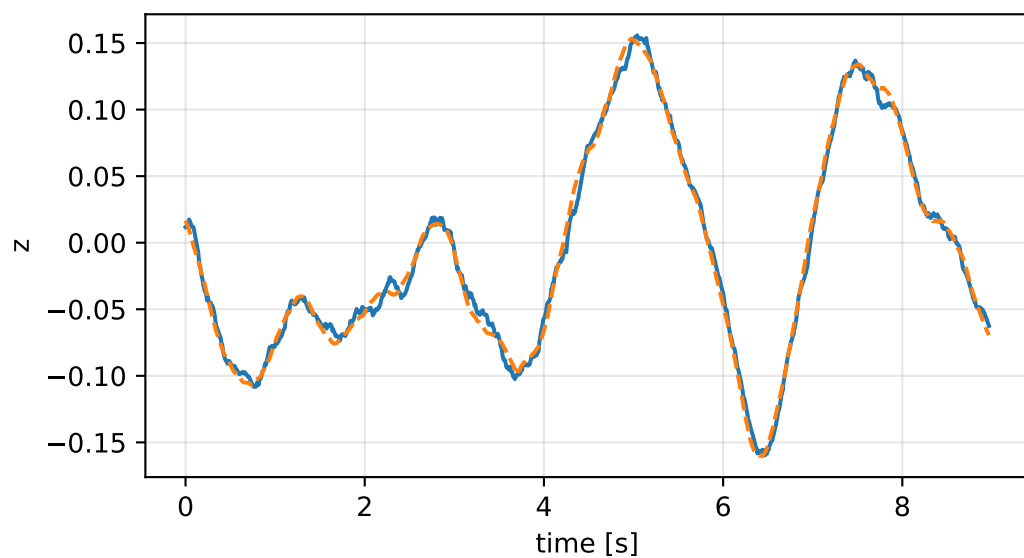
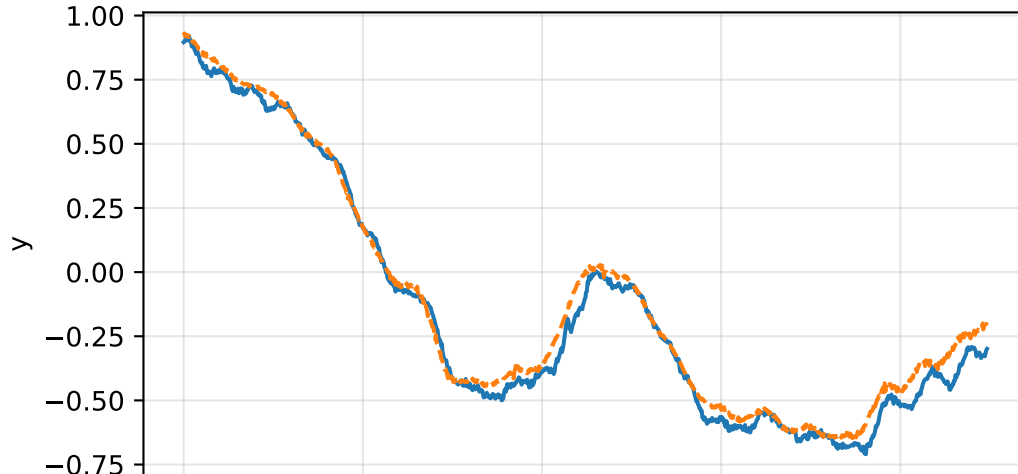
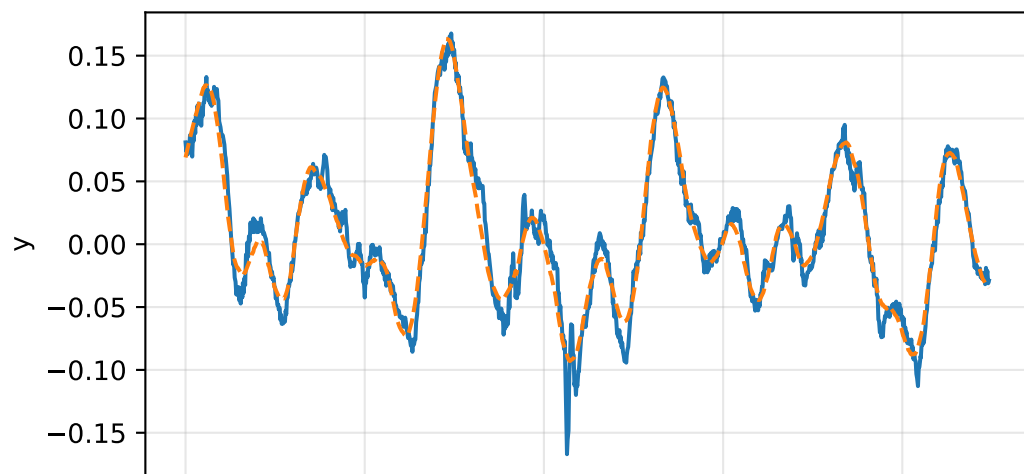
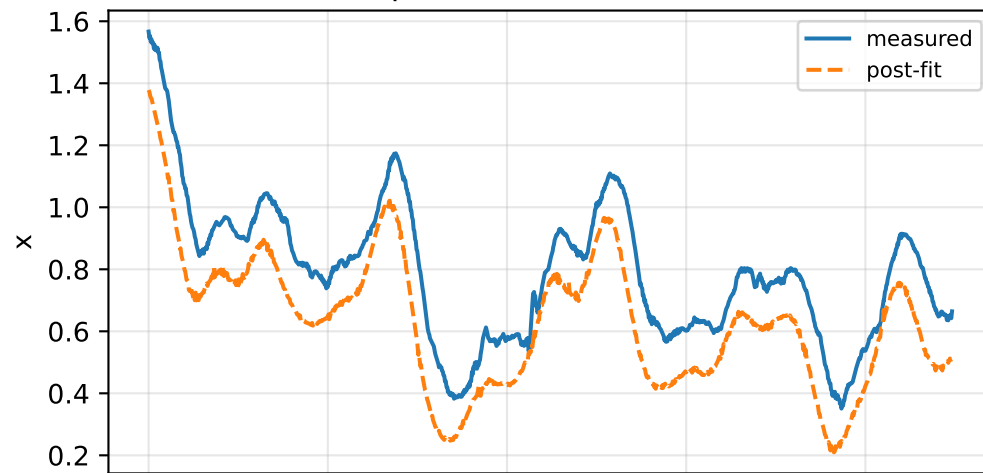


kkt_scale_positive: IMU comparison (diagnostic only)

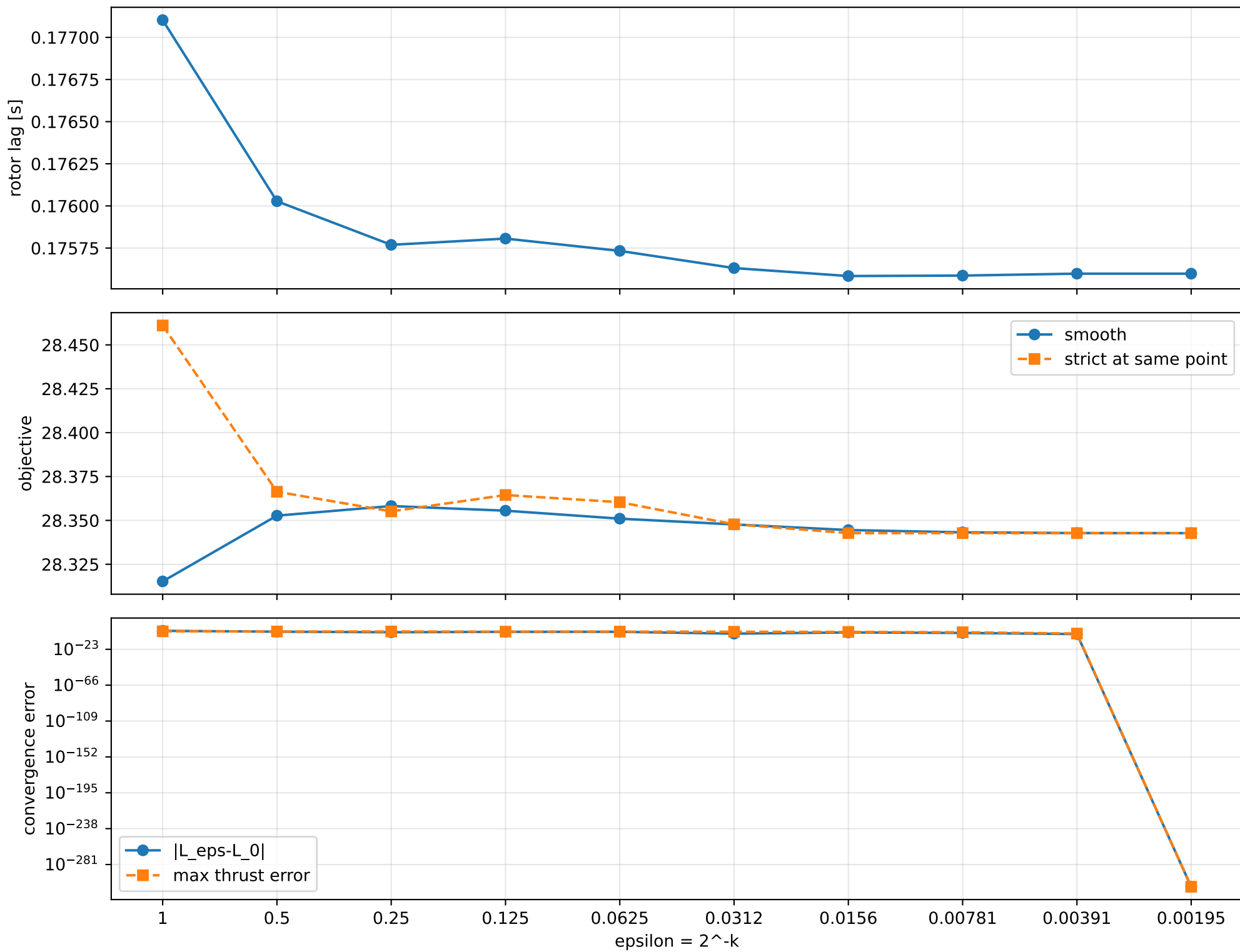
gyro [rad/s]



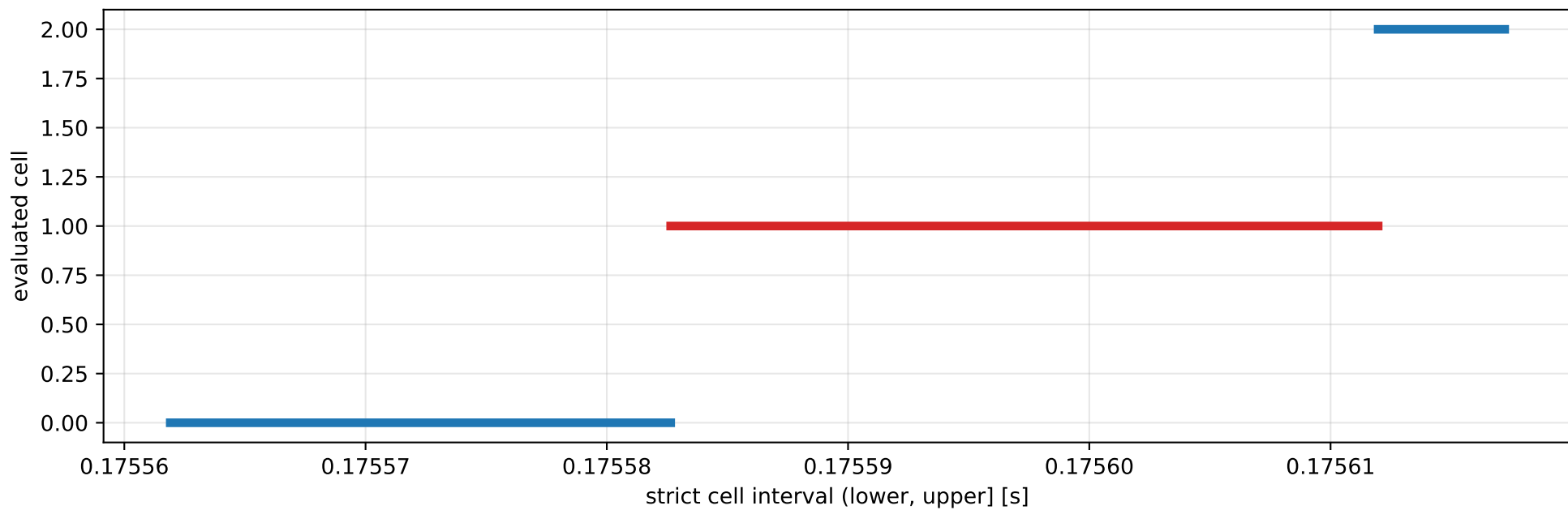
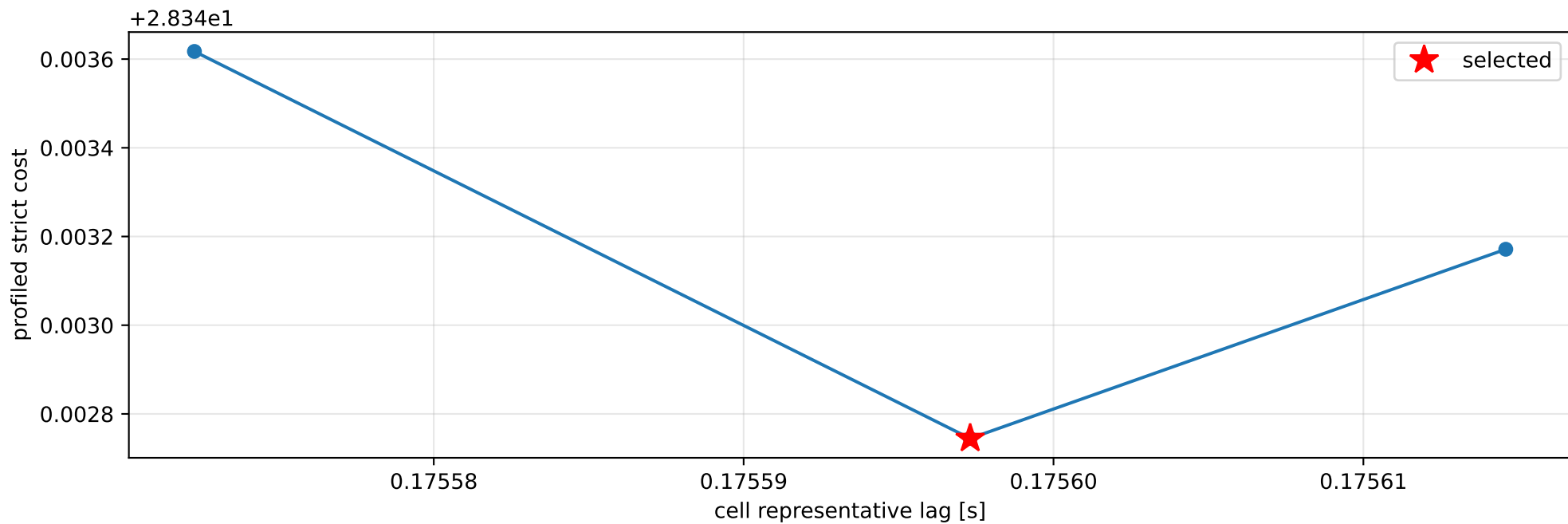
specific force [m/s^2]



kkt_scale_positive: smooth-to-strict continuation



kkt_scale_positive: exact strict-ZOH lag cells



selected (0.175582647, 0.175611973] s; final smooth 0.175597984 s; data boundary=False

kkt_scale_positive: parameter estimate

Case: kkt_scale_positive

status: completed (all stages completed)

mass [kg]: nominal 2.35159759; estimated 5.95395076

inertia [kg m²]:

```
[[ 1.38088922 -0.03064693  0.10745121]
 [-0.03064693  0.65296996 -0.01821479]
 [ 0.10745121 -0.01821479  0.76743359]]
```

CoG body [m]: [-0.01982262 -0.05513055 -0.00503056]

force effectiveness: [2.64671285 2.58999671 1.36647128 1.65924692]

scale-free J/m [m²]:

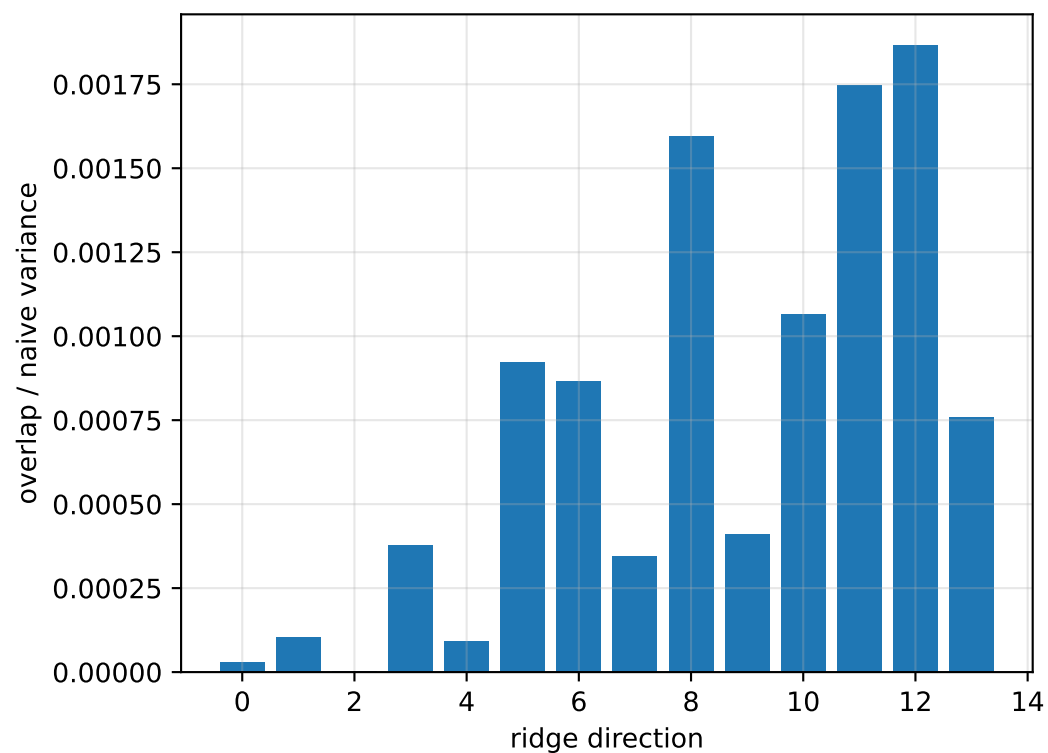
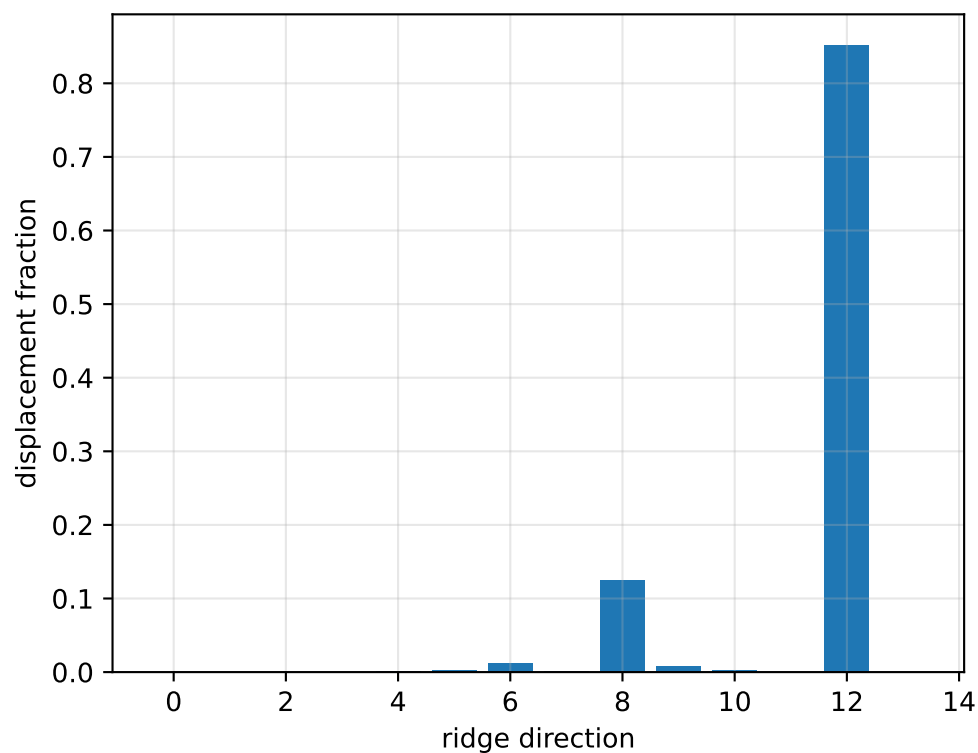
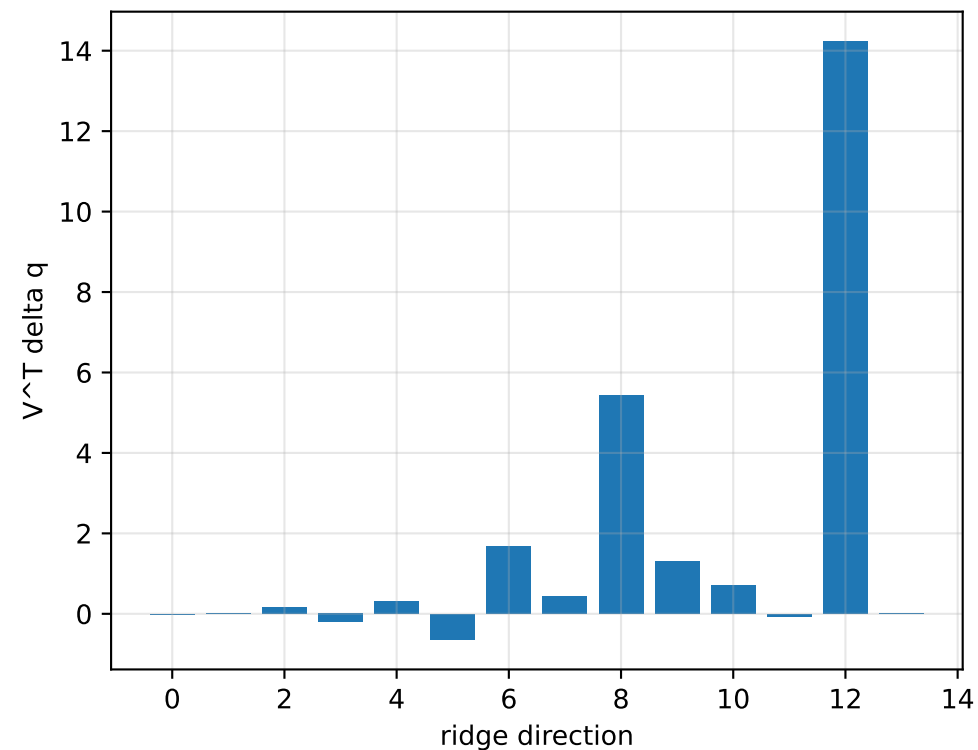
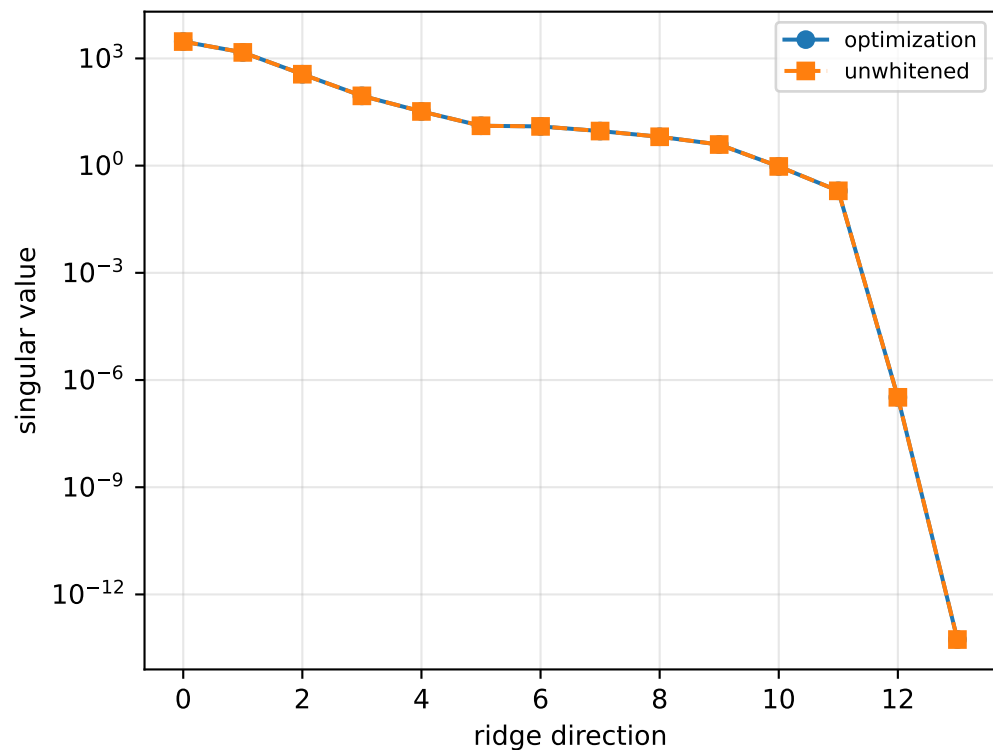
```
[[ 0.23192822 -0.00514733  0.01804704]
 [-0.00514733  0.10967003 -0.00305928]
 [ 0.01804704 -0.00305928  0.12889485]]
```

scale-free f/m: [0.44453052 0.43500472 0.22950665 0.27867999]

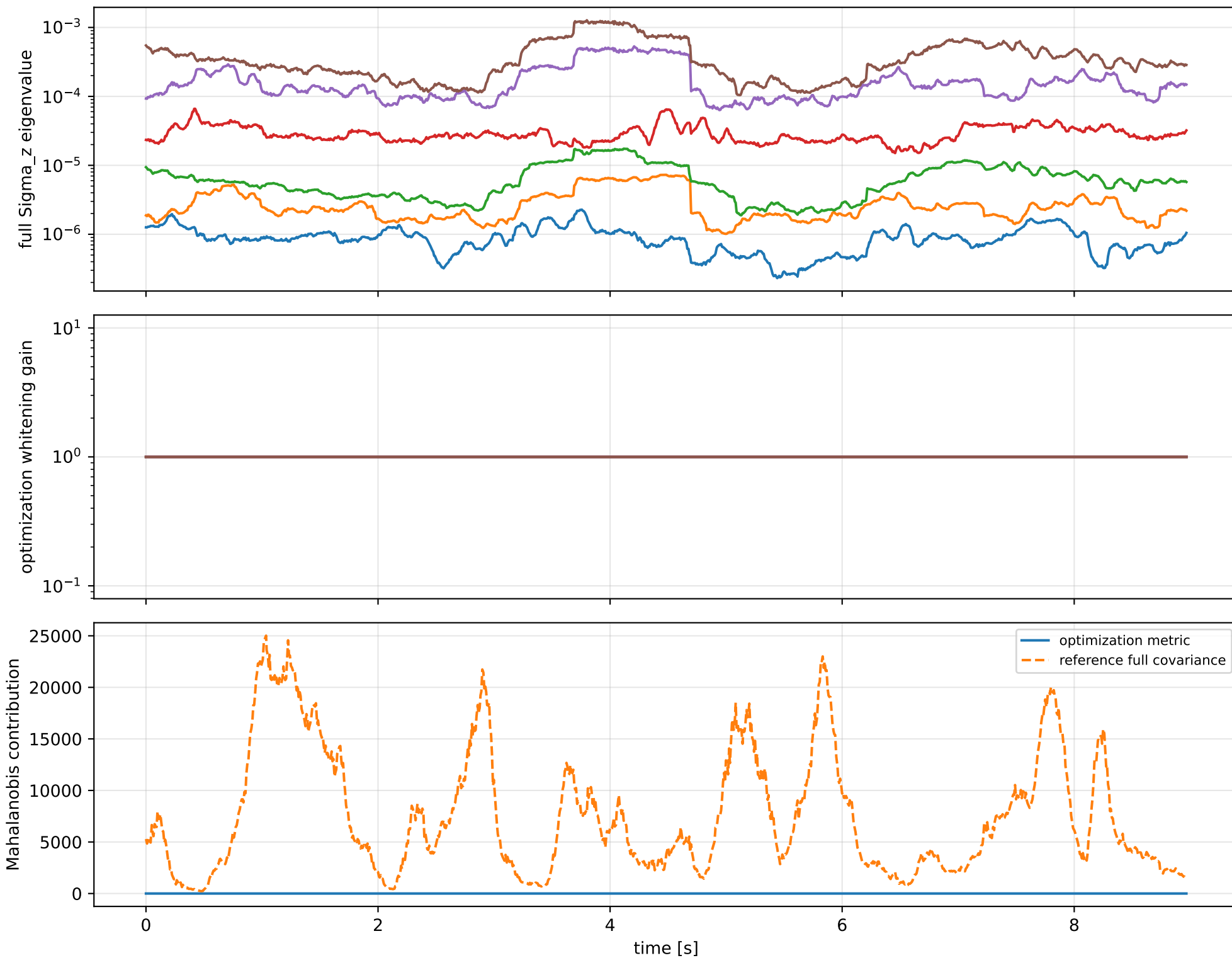
rotor lag cell: (0.175582647, 0.175611973] s

representative: 0.17559731 s

kkt_scale_positive: ridge spectrum (rank 13, nullity 1, $||\mathbf{J}\mathbf{v}||=7.97\text{e-}14$)



kkt_scale_positive: optimization vs reference covariance



kkt_scale_positive: wrench / closure (force max 1.69e-14, torque max 1.11e-16)

